

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

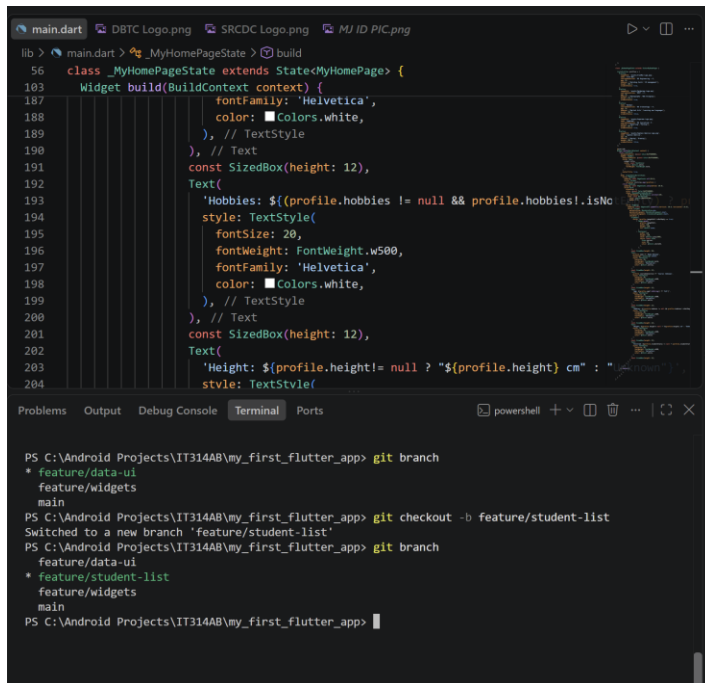
Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch



```
lib > main.dart > MyHomePageState > build
56 class MyHomePageState extends StatelessWidget {
103   Widget build(BuildContext context) {
187     const TextStyle(
188       fontFamily: 'Helvetica',
189       color: Colors.white,
190     ), // Text
191     const SizedBox(height: 12),
192     Text(
193       'Hobbies: ${(profile.hobbies != null && profile.hobbies!.isNotEmpty) ? profile.hobbies!.join(', ') : ''}',
194       style: TextStyle(
195         fontSize: 20,
196         fontWeight: FontWeight.w500,
197         fontFamily: 'Helvetica',
198         color: Colors.white,
199       ), // Text
200     ), // Text
201     const SizedBox(height: 12),
202     Text(
203       'Height: ${(profile.height != null ? "${profile.height} cm" : "Not known")}',
204       style: TextStyle(
205         fontSize: 20,
206         fontWeight: FontWeight.w500,
207         fontFamily: 'Helvetica',
208         color: Colors.white,
209       ), // Text
210     ), // Text
211   )
212 }
213
214 PS C:\Android Projects\IT314AB\my_first_flutter_app> git branch
* feature/data-ui
  feature/widgets
  main
PS C:\Android Projects\IT314AB\my_first_flutter_app> git checkout -b feature/student-list
Switched to a new branch 'feature/student-list'
PS C:\Android Projects\IT314AB\my_first_flutter_app> git branch
* feature/data-ui
  feature/student-list
  feature/widgets
  main
PS C:\Android Projects\IT314AB\my_first_flutter_app>
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart **List**.

Research what a Dart **List** is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

Code:

- Your student list
- At least five student entries

```

class _MyHomePageState extends State<MyHomePage> {
  List<Profile> profiles = [
    Profile(
      imagePath: 'assets/IronMan Logo.png',
      name: 'Iron Man',
      courseAndSection: 'BS Engineering - 4',
      age: 23,
      hobbies: ['Building Suits', 'AI management'],
      height: 185.2,
      studentStatus: true,
    ),
    Profile(
      imagePath: 'assets/Spiderman Logo.png',
      name: 'Spiderman',
      courseAndSection: 'BSIT - 2',
      age: 19,
      hobbies: ['Photography', 'Web Slinging'],
      height: 175.2,
      studentStatus: true,
    ),
    Profile(
      imagePath: 'assets/Batman Logo.png',
      name: 'Batman',
      courseAndSection: 'BS Criminology - 4',
      age: 24,
      hobbies: ['Martial Arts', 'Learning new languages'],
      height: 186.5,
      studentStatus: false,
    ),
    Profile(
      imagePath: 'assets/Superman Logo.png',
      name: 'Superman',
      courseAndSection: 'BA Journalism - 3',
      age: 22,
      hobbies: ['Reporting', 'Farming'],
      height: 187.0,
      studentStatus: true,
    ),
    Profile(
      imagePath: 'assets/Captain America Logo.png',
      name: 'Captain America',
      courseAndSection: 'BS Military Science - 4',
      age: 30,
      hobbies: ['Boxing', 'Drawing'],
      height: 189.0,
      studentStatus: false,
    ),
  ];
}

```

🚩 Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

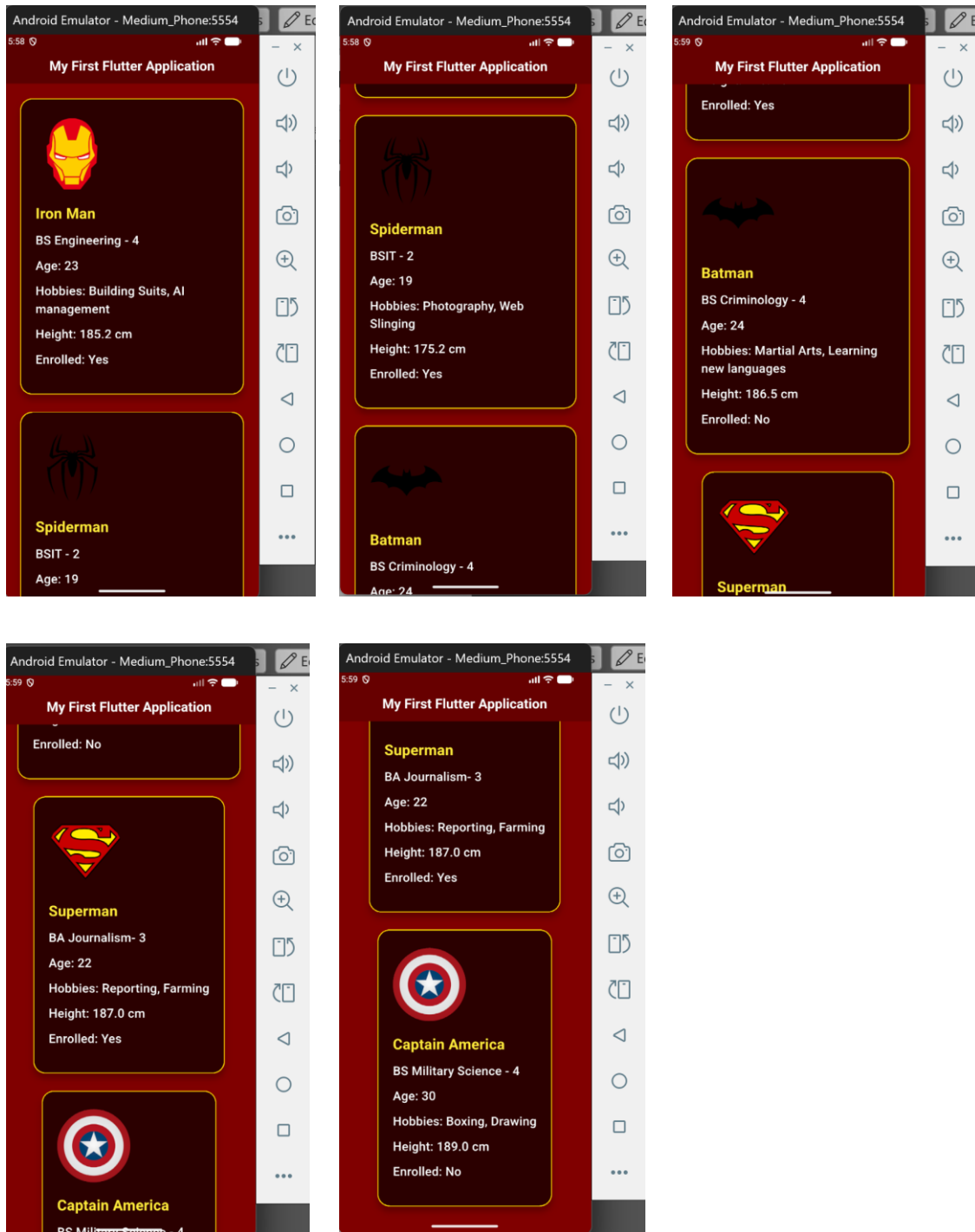
- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

- Your five manually created student cards (do not change your past code yet)



🚩 Flag 3 — Discover **ListView**

Objective: Make your student directory scrollable using **ListView**.

Research Flutter's **ListView**.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

Test

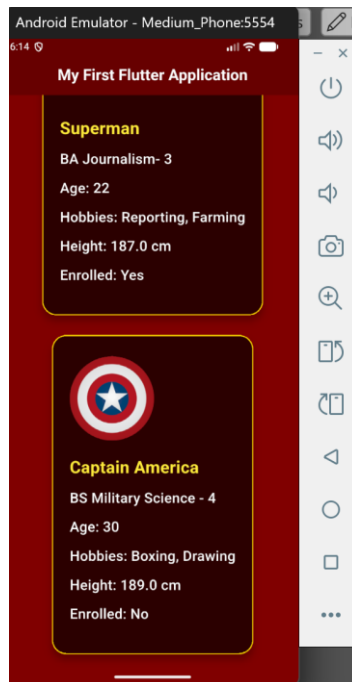
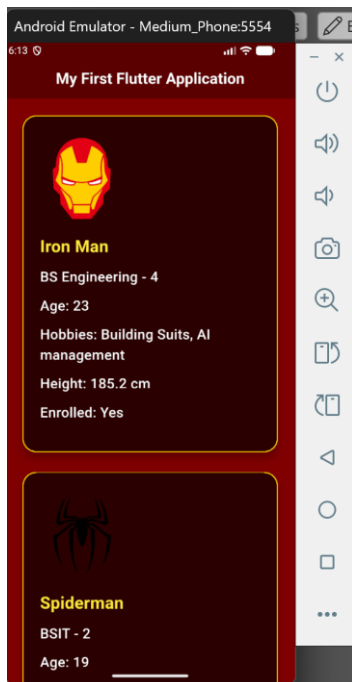
The user should be able to:

- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:

- `ListView` in your code
- Top of the application
- Bottom portion showing later students

```
body: ListView(  
  padding: const EdgeInsets.all(20.0),  
  children: profiles.map((profile) {  
    return Padding(  
      padding: const EdgeInsets.all(10.0),  
      child: Card(  
        child: Column(  
          crossAxisAlignment: CrossAxisAlignment.start,  
          children: [  
            profile.image,   
            profile.name,   
            profile.degree,   
            profile.age,   
            profile.hobbies,   
            profile.height,   
            profile.enrolled,   
          ],  
        ),  
      ),  
    ),  
  ),  
);
```



🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

Test

Add one new student to your list.

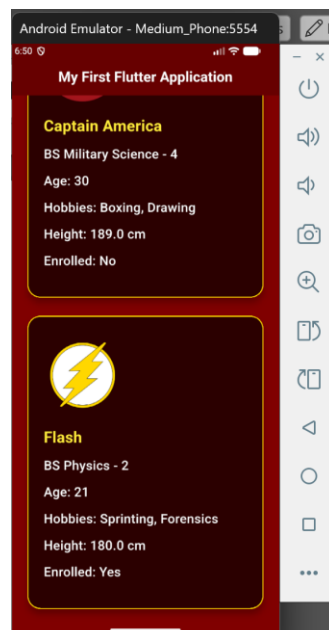
Without creating another card manually, your application should display the new student automatically.

Screenshot:

- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app

```
body: ListView.builder(  
  padding: const EdgeInsets.all(20.0),  
  itemCount: profiles.length,  
  itemBuilder: (context, index){  
    final profile = profiles[index];  
    return Padding(  
      padding: const EdgeInsets.all(10.0),  
      child: Card(  
        child: Column(  
          crossAxisAlignment: CrossAxisAlignment.start,  
          children: [  
            Image.asset(profile.imagePath, width: 50, height: 50),  
            Text(profile.name),  
            Text(profile.courseAndSection),  
            Text('Age: ${profile.age}'),  
            Text('Hobbies: ${profile.hobbies.join(', ')}'),  
            Text('Height: ${profile.height} cm'),  
            Text('Enrolled: ${profile.enrolled}'),  
          ],  
        ),  
      ),  
    );  
  },  
);
```

```
List<Profile> profiles = [  
  Profile(  
    imagePath: 'assets/IronMan logo.png',  
    name: 'Iron Man',  
    courseAndSection: 'BS Engineering - 4',  
    age: 23,  
    hobbies: ('Building Suits', 'AI management'),  
    height: 185.2,  
    studentStatus: true,  
  ),  
  Profile(  
    imagePath: 'assets/Spiderman logo.png',  
    name: 'Spiderman',  
    courseAndSection: 'BSIT - 2',  
    age: 19,  
    hobbies: ('Photography', 'Web Slinging'),  
    height: 175.2,  
    studentStatus: true,  
  ),  
  Profile(  
    imagePath: 'assets/Batman logo.png',  
    name: 'Batman',  
    courseAndSection: 'BS Criminology - 4',  
    age: 24,  
    hobbies: ('Martial Arts', 'Learning new languages'),  
    height: 180.5,  
    studentStatus: false,  
  ),  
  Profile(  
    imagePath: 'assets/Superman logo.png',  
    name: 'Superman',  
    courseAndSection: 'BA Journalism - 3',  
    age: 22,  
    hobbies: ('Reporting', 'Farming'),  
    height: 187.8,  
    studentStatus: true,  
  ),  
  Profile(  
    imagePath: 'assets/Captain America logo.png',  
    name: 'Captain America',  
    courseAndSection: 'BS Military Science - 4',  
    age: 30,  
    hobbies: ('Boxing', 'Drawing'),  
    height: 189.0,  
    studentStatus: false,  
  ),  
  Profile(  
    imagePath: 'assets/Flash logo.png',  
    name: 'Flash',  
    courseAndSection: 'BS Physics - 2',  
    age: 21,  
    hobbies: ('Sprinting', 'Forensics'),  
    height: 180.0,  
    studentStatus: true,  
  ),  
];
```



🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

- Student ID
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

- Updated student data
- Student Card displaying the additional fields

```
class Profile{
    String? name;
    String? courseAndsection;
    int? age;
    Set<String>? hobbies;
    double? height;
    bool? studentStatus;
    String? imagePath;
    String? studentId;
    String? email;
    String? phoneNumber;

    Profile({
        this.name,
        this.courseAndsection,
        this.age,
        this.hobbies,
        this.height,
        this.studentStatus,
        this.imagePath,
        this.studentId,
        this.email,
        this.phoneNumber
    });
}
```

```
Profile(
    imagePath: 'assets/Superman Logo.png',
    name: 'Superman',
    courseAndsection: 'BA Journalism- 3',
    age: 22,
    hobbies: {'Reporting', 'Farming'},
    height: 187.0,
    studentId: '2020-0004',
    email: 'superman@dbtc.edu.ph',
    phoneNumber: '09204567890',
    studentStatus: true,
),
```



🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

```
[]
```

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

- "No students found."
- "Student list is empty."

Test

- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

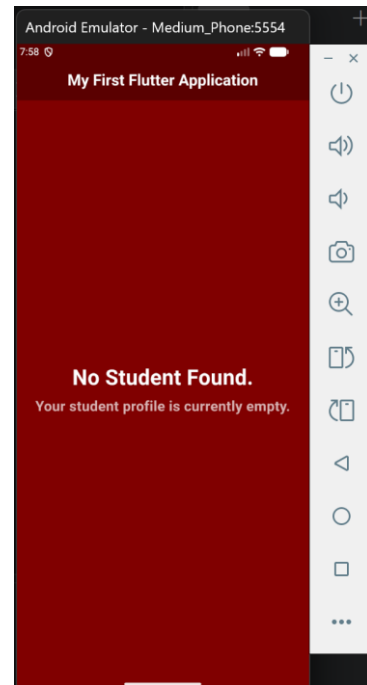
- Empty student list
- Empty-state UI
- Application still functioning correctly

```
class _MyHomePageState extends State<MyHomePage> {  
  
  List<Profile> profiles = [  
  
  ];
```

```

), // AppBar
body: profiles.isEmpty ?
const Center(
  child: Column(
    mainAxisAlignment: MainAxisAlignment.center,
    crossAxisAlignment: CrossAxisAlignment.center,
    children: [
      Text('No Student Found.',
        style: TextStyle(
          fontSize: 30,
          fontWeight: FontWeight.bold,
          fontFamily: 'Helvetica',
          color: Colors.white,)), // TextStyle // Text
      SizedBox(height: 5),
      Text('Your student profile is currently empty.',
        style: TextStyle(
          fontSize: 20,
          fontWeight: FontWeight.bold,
          fontFamily: 'Helvetica',
          color: Colors.white70,)), // TextStyle // Text
      SizedBox(height: 20)
    ]
  ) // Column
) // Center

```



🚩 Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a [List](#).

Sort your students alphabetically by name.

Test

Before sorting:

- Maria
- John
- Aaron
- Sophia

After sorting:

- Aaron
- John
- Maria

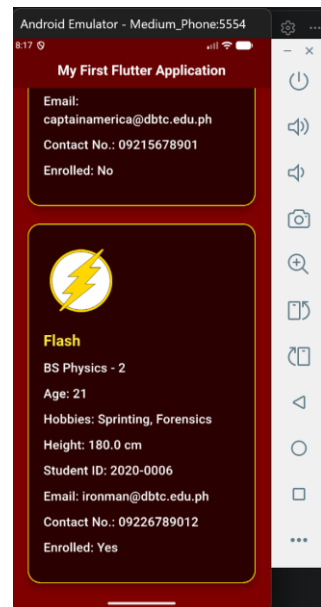
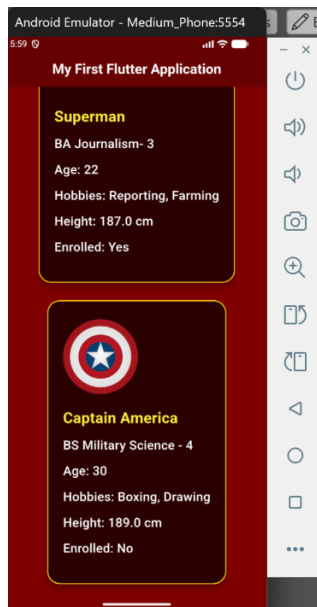
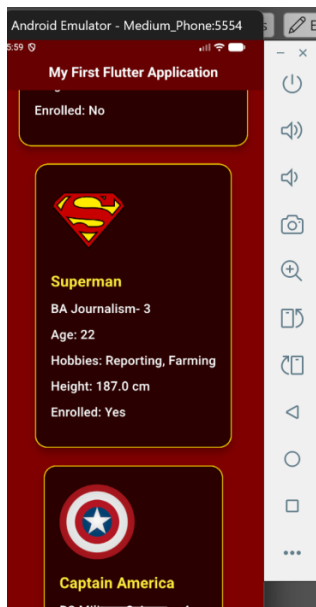
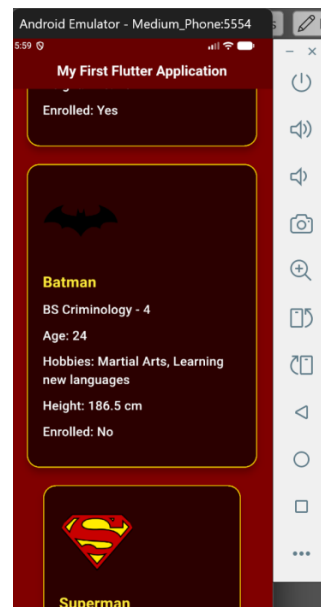
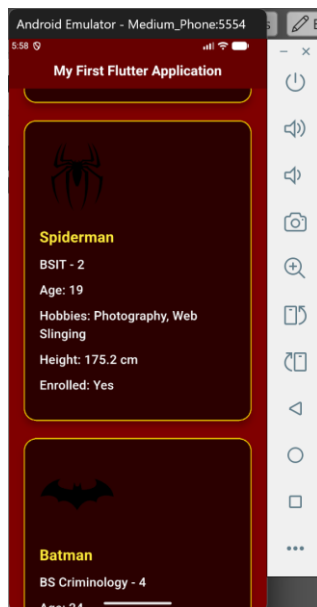
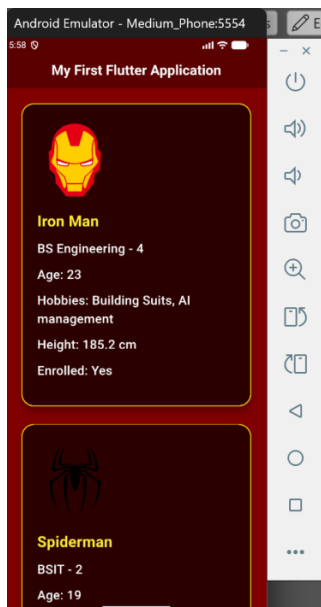
- Sophia

Your UI should immediately reflect the new order.

Screenshot:

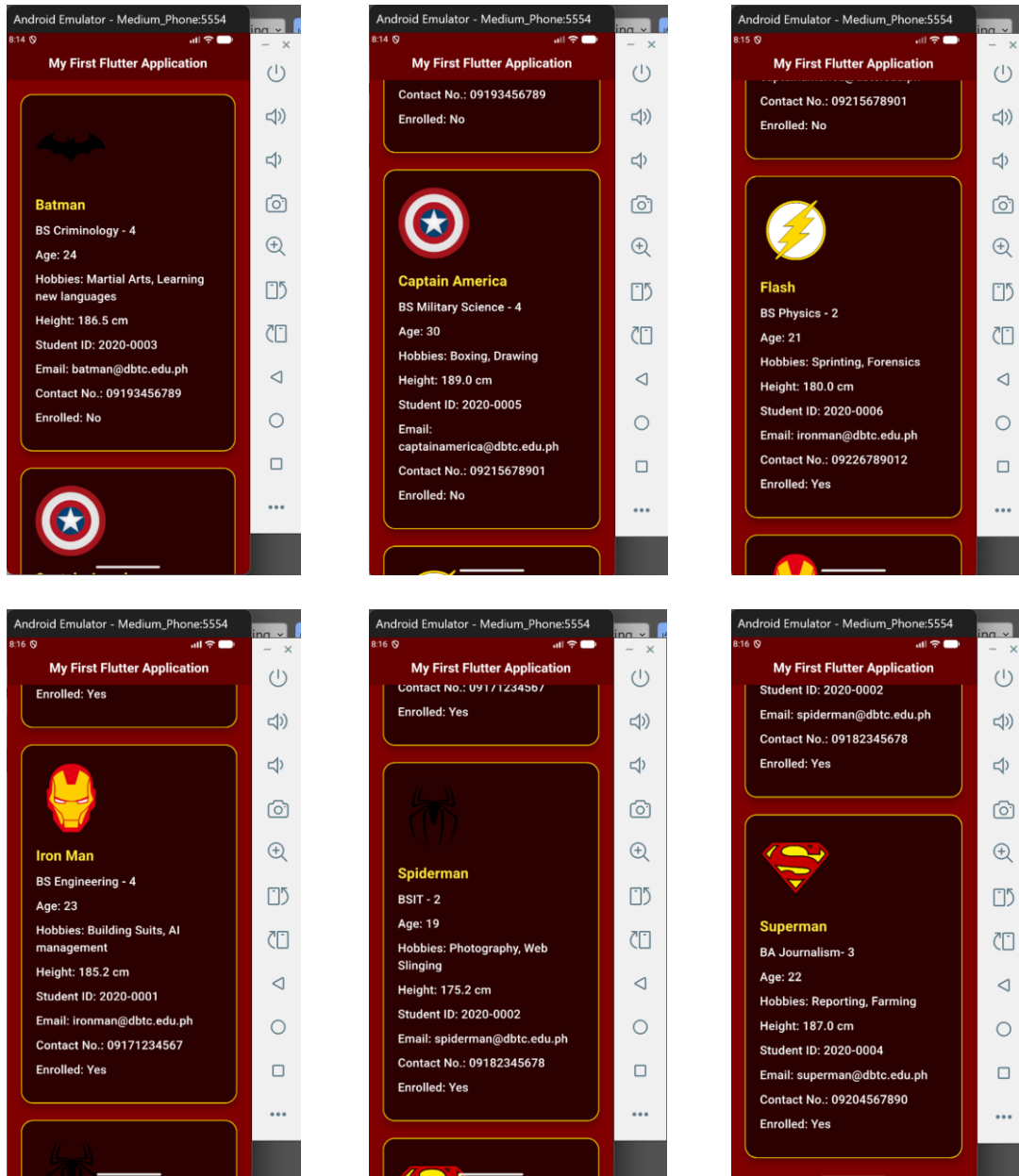
- Student list before or after sorting
- Application displaying the sorted order

Before sorting:



After Sorting:

```
@override  
Widget build(BuildContext context) {  
  profiles.sort((a, b) => (a.name ?? '').compareTo(b.name ?? ''));  
  return Scaffold(  
    backgroundColor: const Color(0xFF800000),  
  );  
}
```



Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

feature/student-list

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
MJ_ARANZADO@LAPTOP-U8N5E77G MINGW64 /c/Android Projects/IT314AB (feature/student-list)
$ git commit -m "feat: added a scrollable list view using ListView.builder, added new student fields, an empty state handler, and a sorted name arrangement"
[feature/student-list 5ca9caa] feat: added a scrollable list view using ListView.builder, added new student fields, an empty state handler, and a sorted name arrangement
2 files changed, 97 insertions(+), 8 deletions(-)
create mode 100644 my_first_flutter_app/assets/Flash Logo.png

MJ_ARANZADO@LAPTOP-U8N5E77G MINGW64 /c/Android Projects/IT314AB (feature/student-list)
$ git push origin -u feature/student-list
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 18 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 578.54 KiB | 13.15 MiB/s, done.
Total 7 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
remote:
remote: Create a pull request for 'feature/student-list' on GitHub by visiting:
remote:   https://github.com/EmJayAA/IT314AB_Aranzado/pull/new/feature/student-list
remote:
To https://github.com/EmJayAA/IT314AB_Aranzado.git
 * [new branch]      feature/student-list -> feature/student-list
branch 'feature/student-list' set up to track 'origin/feature/student-list'.

MJ_ARANZADO@LAPTOP-U8N5E77G MINGW64 /c/Android Projects/IT314AB (feature/student-list)
$
```

IT314AB_Aranzado Public

feature/student-list had recent pushes 57 seconds ago [Compare & pull request](#)

feature/student-list 4 Branches 0 Tags [Add file](#) [Code](#)

This branch is 5 commits ahead of main. [Contribute](#)

EmJayAA	feat: added a scrollable list view using ListView.builder, added new ...	5ca9caa · 1 minute ago	8 Commits
my_first_flutter_app	feat: added a scrollable list view using ListView.builder, added...	1 minute ago	
IT314AB-CaptureTheFlag2_Aranzado.pdf	pdf: Submit Capture the Flag 2 Activity	last week	
IT314AB-CaptureTheFlag3_Aranzado.pdf	pdf: Submit Capture the Flag 3 Activity	5 days ago	
IT314AB-CaptureTheFlag_Aranzado.pdf	pdf: Submit Capture the Flag Activity	2 weeks ago	
IT314AB_Aranzado.pdf	pdf: Submit Assignment #1	2 weeks ago	